

Inventory Management System (IMS) – Project Report

1. Project Overview

Title:

Multi-Tenant Subscription-Based Inventory Management System

Objective:

To design and develop a scalable SaaS-based inventory management system that enables multiple companies to manage their products, stock, and team members efficiently under subscription-based plans.

Core Idea:

A centralized platform serving multiple businesses with separate data, controlled access, and subscription-based usage.

2. Key Features

- Multi-tenant architecture (multiple companies with isolated data)
 - Secure authentication using JWT
 - Product and category management
 - Real-time inventory tracking
 - Low stock alerts
 - Role-based access control (Admin, Staff)
 - Subscription-based feature restrictions
 - Structured and functional dashboard interface
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3. System Architecture

High-Level Architecture

Frontend (React / Next.js)

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Backend API (Node.js / Express)

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Database (MongoDB / MySQL)

Request Flow

User Action → Frontend → API → Backend → Database → Response → UI Update

Authentication Flow

Login → Verify credentials → Generate JWT

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Frontend stores token

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Each request includes token

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Backend verifies token and authorizes access

4. Tech Stack

Frontend

- React.js / Next.js
- Tailwind CSS
- Axios

Backend

- Node.js
- Express.js
- JWT Authentication
- Bcrypt

Database

- MongoDB or MySQL

Deployment

- Frontend: Vercel
 - Backend: Render or Railway
 - Database: MongoDB Atlas
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5. Database Design

Tables / Collections

Companies

- id
- name
- email
- plan
- subscription_start
- subscription_end

Users

- id
- company_id
- name
- email
- password
- role

Products

- id
- company_id
- name
- sku
- category_id
- price
- stock
- reserved_stock

Categories

- id
- company_id

- name

Key Concept:

Each table contains a `company_id` field to ensure data isolation in a multi-tenant environment.

6. Core Modules and Working Flow

Authentication Module

- Company registration and admin creation
- Secure login using JWT

Product Management

- Add, edit, delete, and view products
- Products are linked to a specific company

Category Management

- Create and manage product categories

Inventory Management

Stock Types:

- Available Stock
- Reserved Stock
- Sold Stock

Workflow:

Order placed → Stock reserved

Order shipped → Stock deducted

Order cancelled → Stock restored

Dashboard Module

- Displays inventory insights and key metrics

User Management

- Admin can add, update, or remove users
 - Role-based permissions (Admin and Staff)
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7. Subscription System

Plans

Basic Plan

- Maximum 50 products
- 1 user
- Limited features

Pro Plan

- Maximum 500 products
- Up to 5 users
- Additional features such as alerts

Business Plan

- Unlimited products
 - Unlimited users
 - Full feature access
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Subscription Workflow

Register → Select Plan → Store Plan in Database

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Middleware validates plan before allowing operations

Restrictions

- Product limit
 - User limit
 - Feature access control
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8. API Design

Authentication

- POST /register
- POST /login

Products

- GET /products
- POST /products
- PUT /products/:id
- DELETE /products/:id

Categories

- GET /categories
- POST /categories

Dashboard

- GET /dashboard

Subscription

- GET /plans
 - POST /subscribe
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9. Frontend Pages and Functional Structure

Login Page

- Email and password input
 - Login button
 - Link to registration
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Register Page

- Company name
 - Email
 - Password
 - Plan selection
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Dashboard Page

Contains:

- Summary cards (Total Products, Categories, Low Stock, Inventory Value)
 - Low stock table (Name, SKU, Stock, Status)
 - Inventory trend chart
 - Recent activity section
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Products Page

Contains:

- Search functionality
- Filters (category and stock status)
- Add Product button

Table Columns:

- Product Name
 - SKU
 - Category
 - Price
 - Stock
 - Status
 - Actions (Edit/Delete)
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Add / Edit Product Page

Form Fields:

- Product Name
- SKU
- Category
- Price
- Stock Quantity
- Low Stock Threshold
- Image Upload

Categories Page

- Add category functionality
- Category list
- Product count per category

Users Page

- User list with roles
- Add or edit users

Subscription Page

- Current plan details
- Usage tracking (e.g., products used vs limit)
- Plan comparison
- Upgrade option

10. End-to-End Workflow

Customer places order on external platform

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Admin receives order notification

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Stock is reserved

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Order is shipped and stock is deducted

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Dashboard reflects updated inventory

11. Security and Best Practices

- Password hashing using bcrypt
- JWT-based authentication

- Role-based authorization
 - Input validation
 - Structured error handling
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12. Future Enhancements

- Integration with external platforms such as Amazon and Shopify
- Automated inventory synchronization
- Payment gateway integration
- Advanced analytics and reporting
- Mobile application support